

Lab 2: Agile Backlog Creation & Sprint Simulation in Jira

Project: Automated Rubric Assignment Evaluator (ARAE)

Jira Space: Automated Rubric Assignment Evaluator | Board: ARAE board

1. Project Setup

The Jira Scrum project "Automated Rubric Assignment Evaluator" (ARAE) was created and the ARAE board was initialized with an empty backlog, ready for Epics and User Stories to be added.

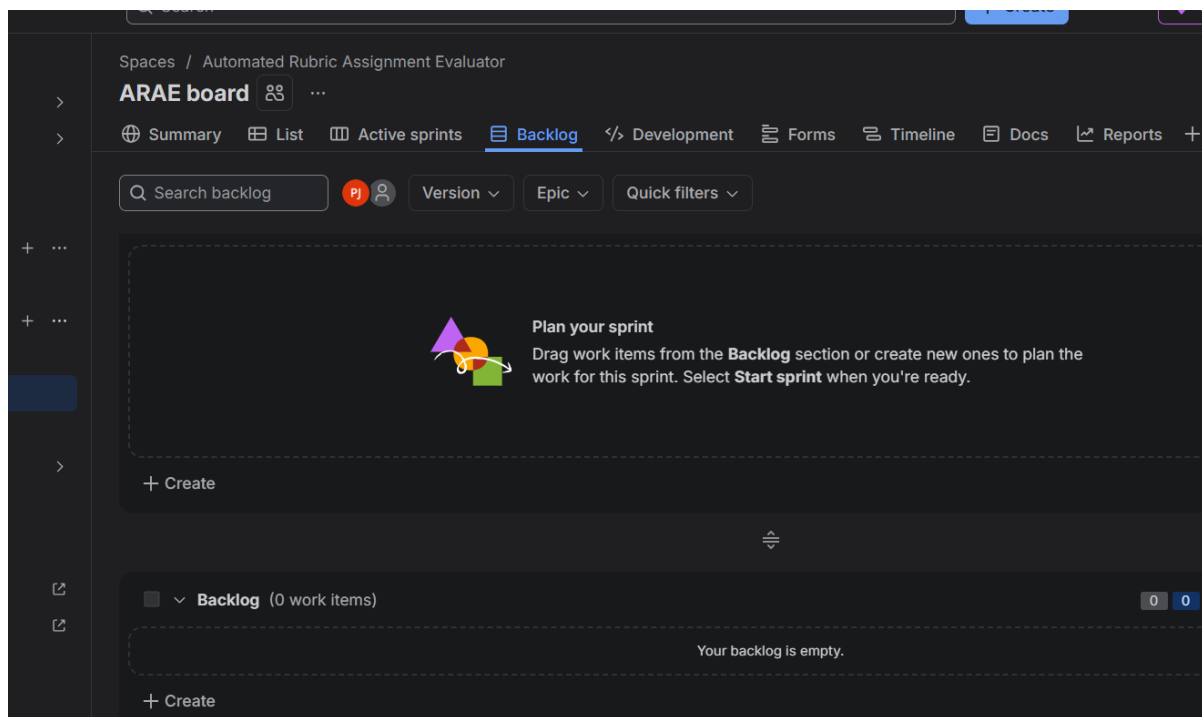


Figure 1: Newly created ARAE board with an empty backlog

2. Create Epics

The Lab 1 functional requirements were broken down into the following major functional themes (Epics): Project Submission Management, Automated Project Testing, Automated Rubric Evaluation, Evaluation Report Management, and Peer Review Management.

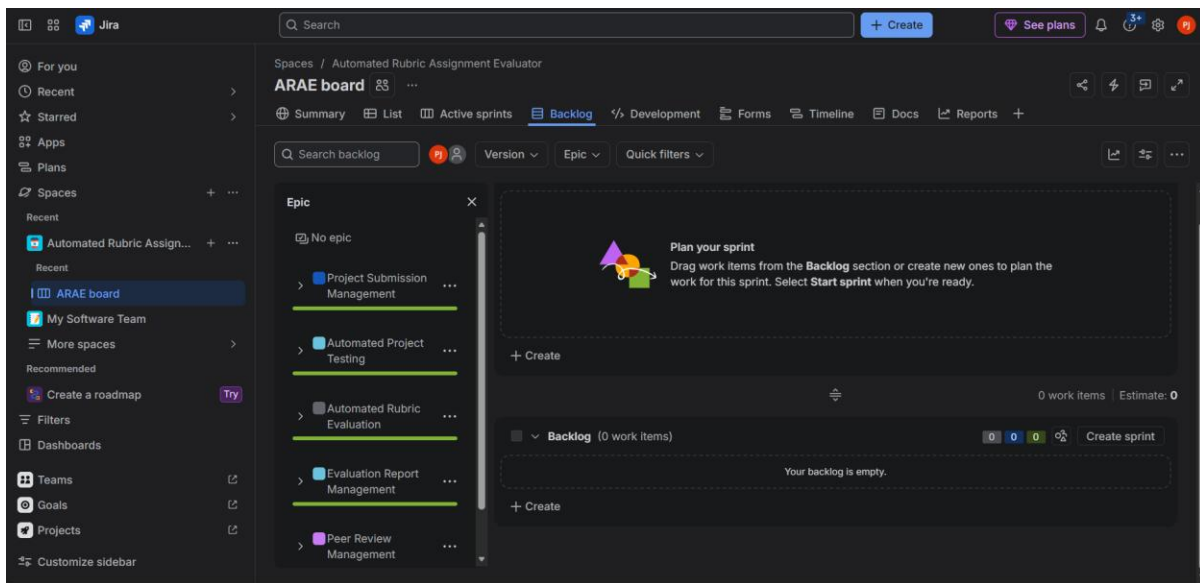


Figure 2: Epics created on the ARAE board

3. Write User Stories & Populate Backlog

For each Epic, related User Stories were written in the standard format ("As a [user role], I want [goal], so that [benefit]") and added as work items in the backlog, grouped under their parent Epics.

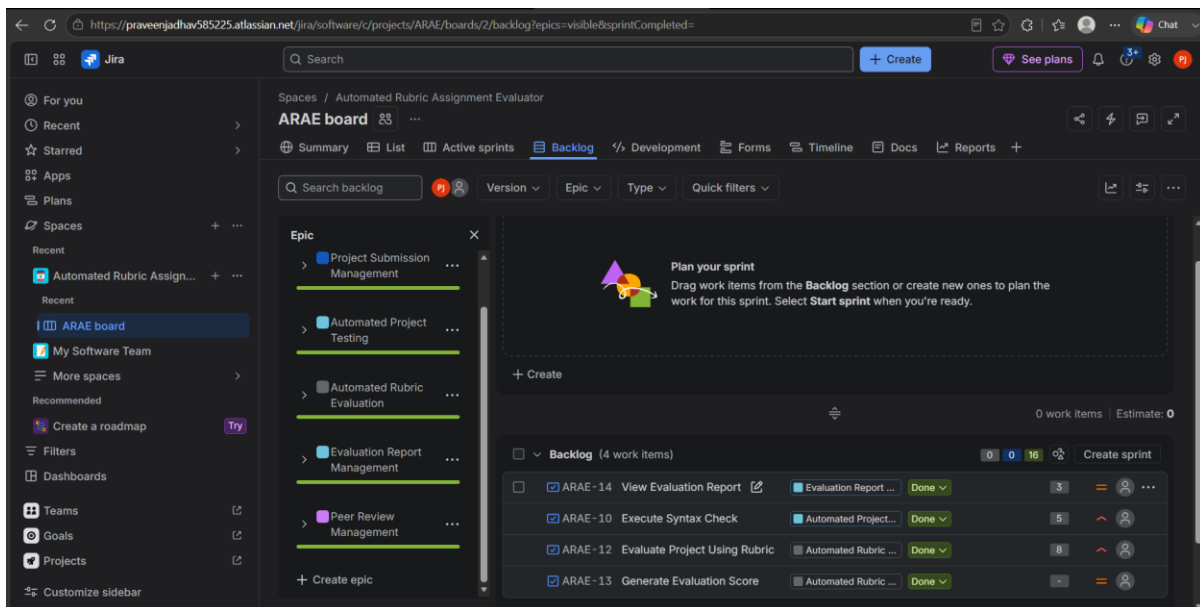


Figure 3: Backlog populated with User Stories grouped under their Epics

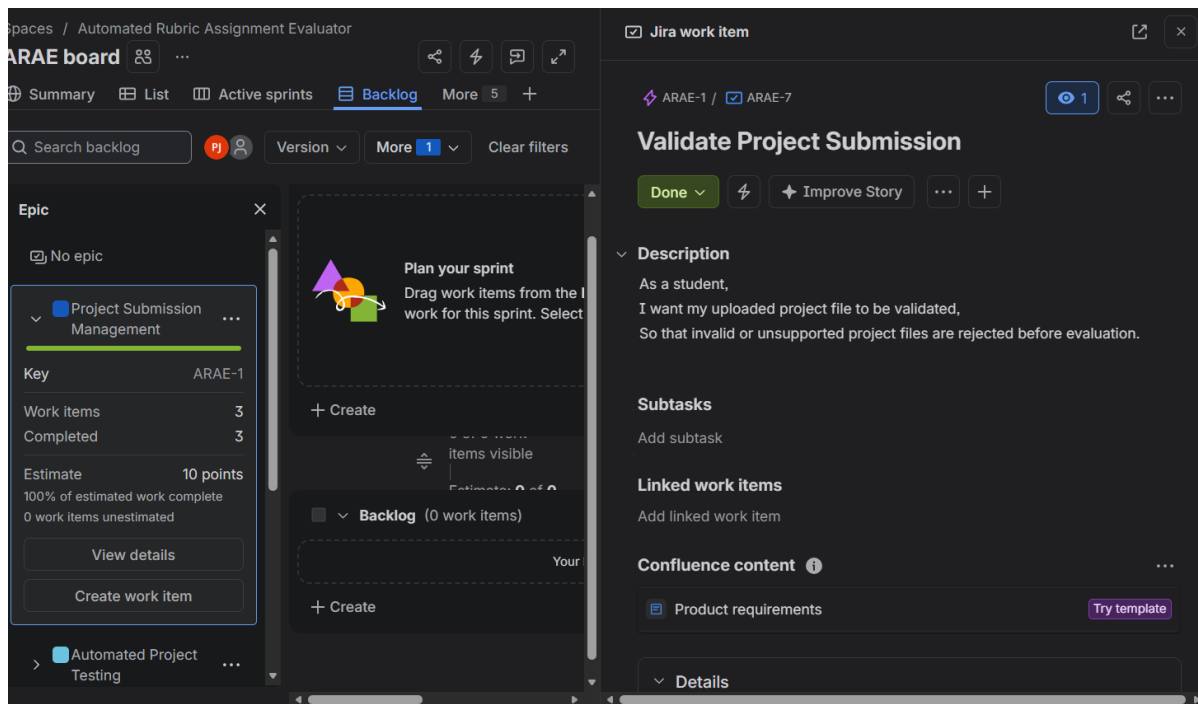


Figure 4: User Story ARAE-7 – "Validate Project Submission", written in the As a / I want / So that format

4. Prioritize the Backlog & Assign Story Points

Each User Story was assigned a priority level (High/Medium/Low) based on user impact, and story points were estimated using the Fibonacci sequence (reflecting complexity, effort, and uncertainty) via a Planning Poker style discussion.

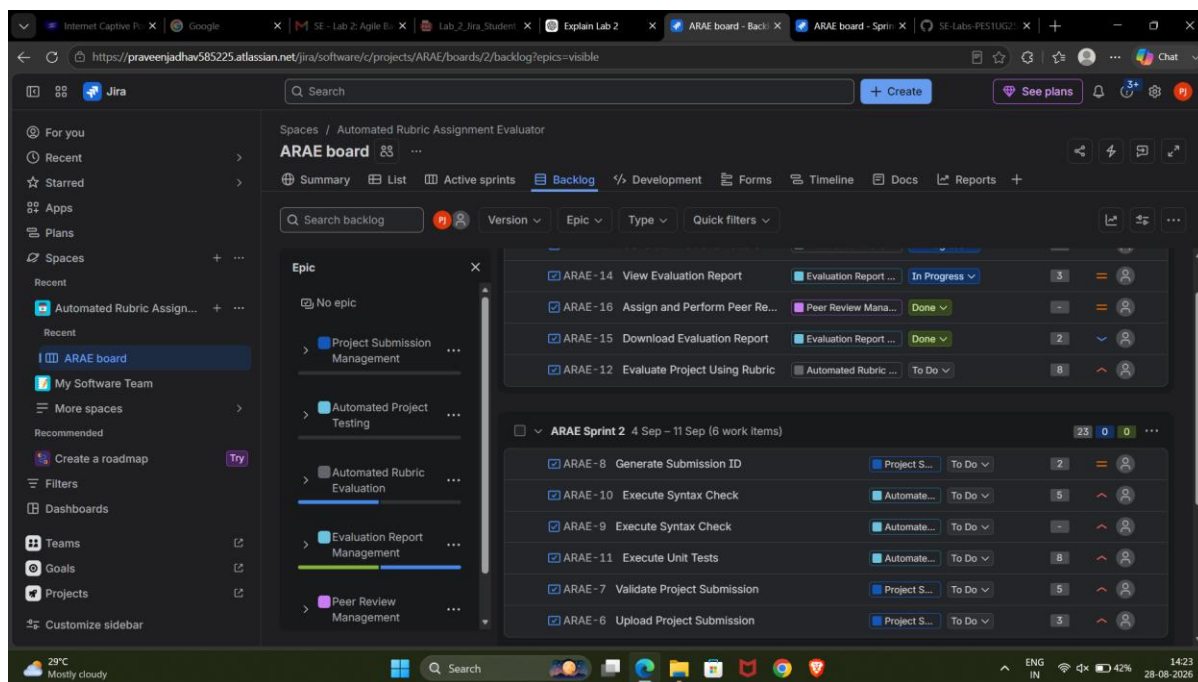


Figure 5: Backlog showing Sprint 2 items with priorities and story point estimates

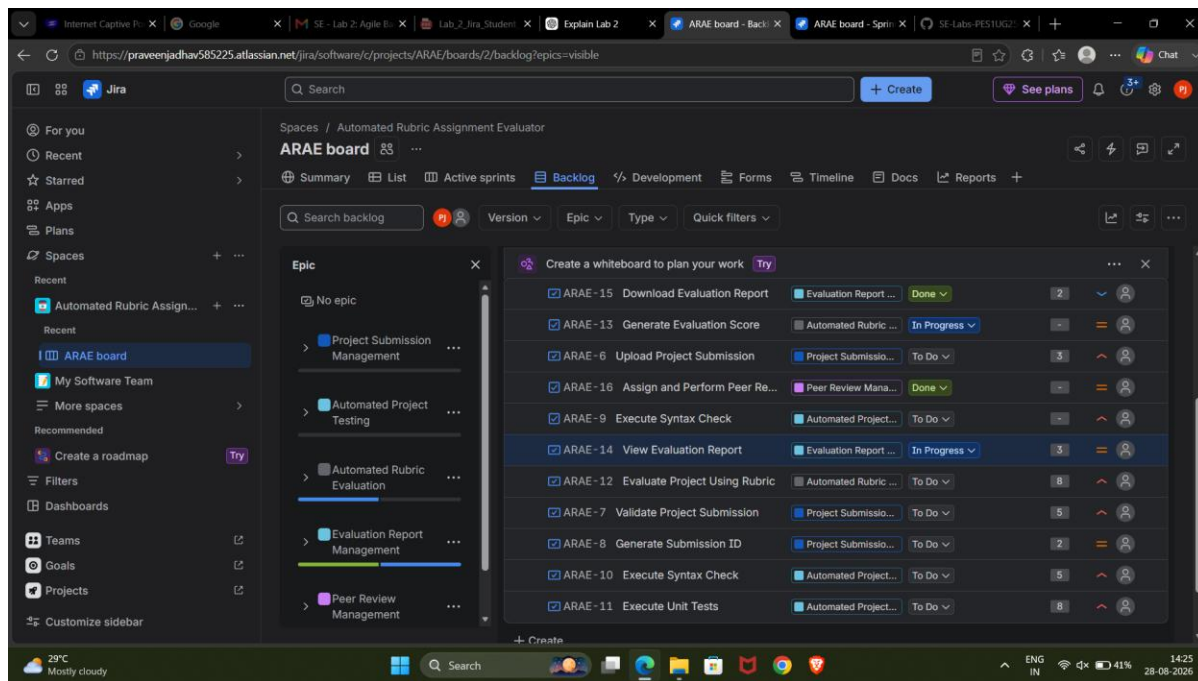


Figure 6: Backlog items with priority flags and story points (continued)

5. Run the Sprint

Sprint 1 was created with a one-week duration, selected User Stories were dragged into the sprint, and the sprint was started. Task progress was then simulated by moving items across the To Do → In Progress → Done columns of the Scrum/Kanban board.

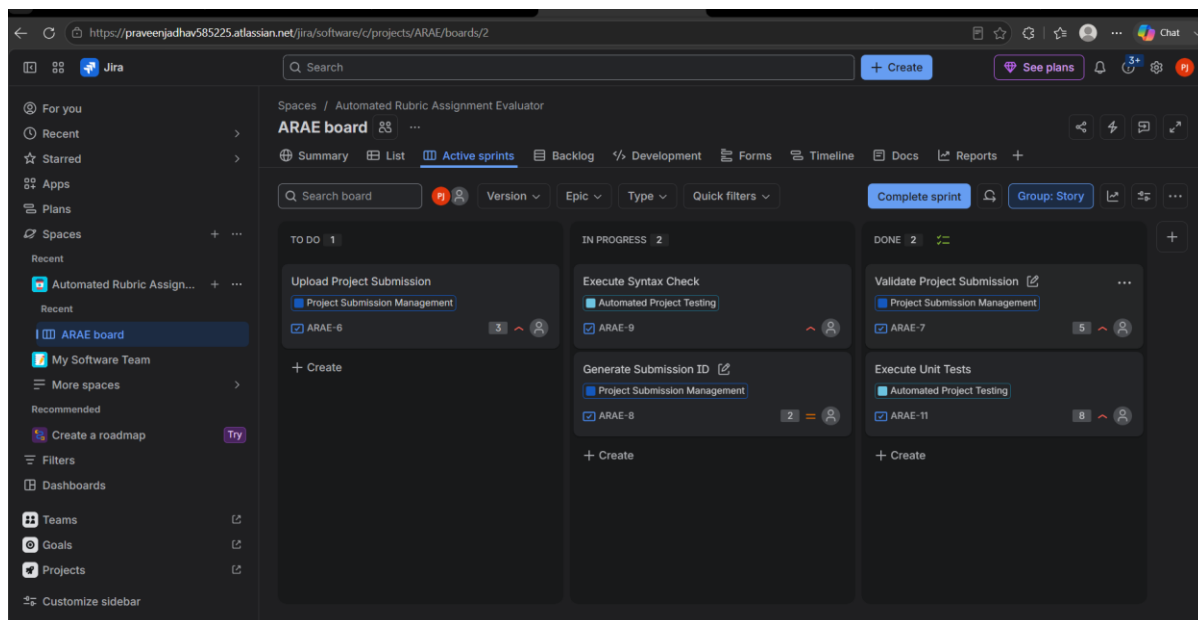


Figure 7: Active sprint board – tasks distributed across To Do, In Progress and Done

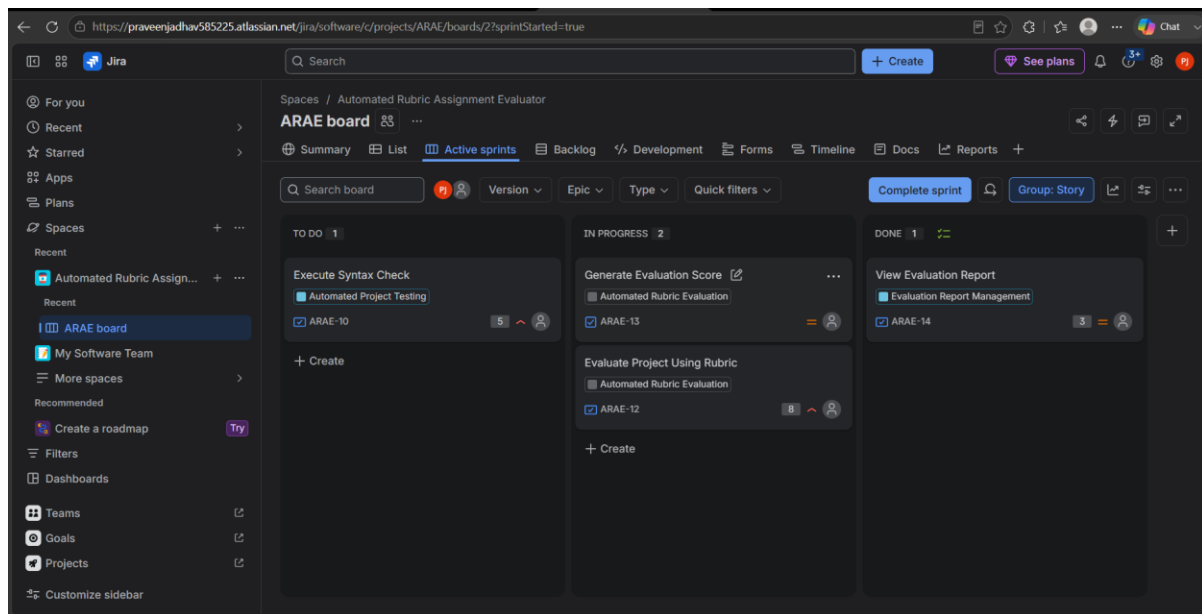


Figure 8: Active sprint board – progress continues as tasks move toward completion

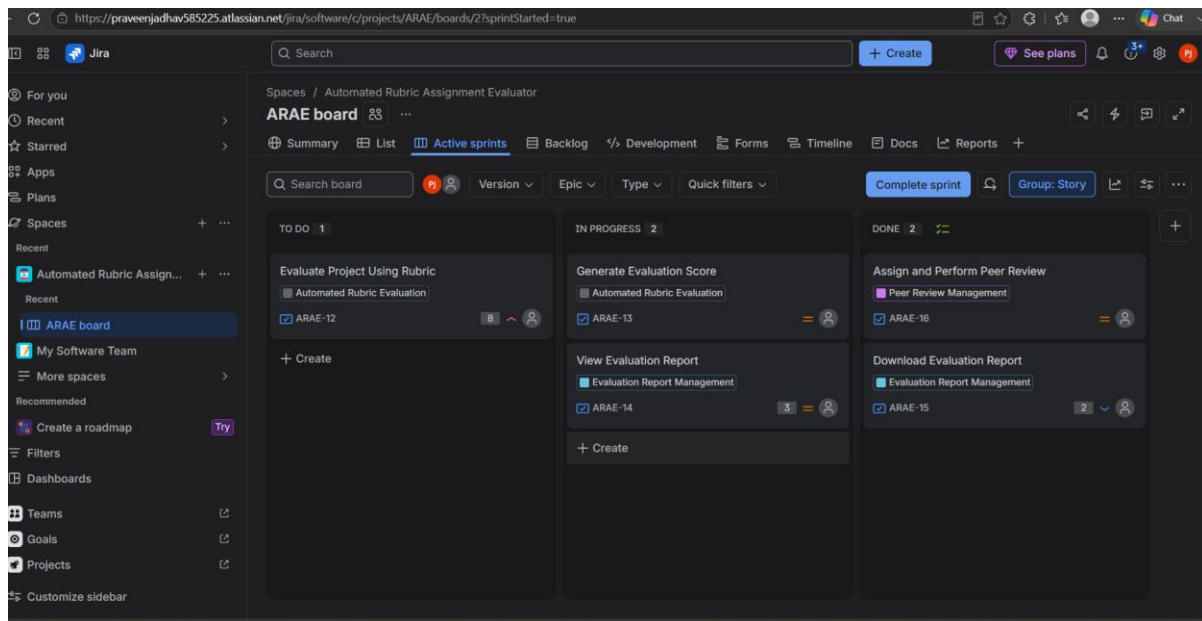


Figure 9: Active sprint board – further tasks marked Done

6. Burndown Chart

The Burndown Chart under Reports was used to monitor story points completed versus the ideal guideline across the sprint duration.

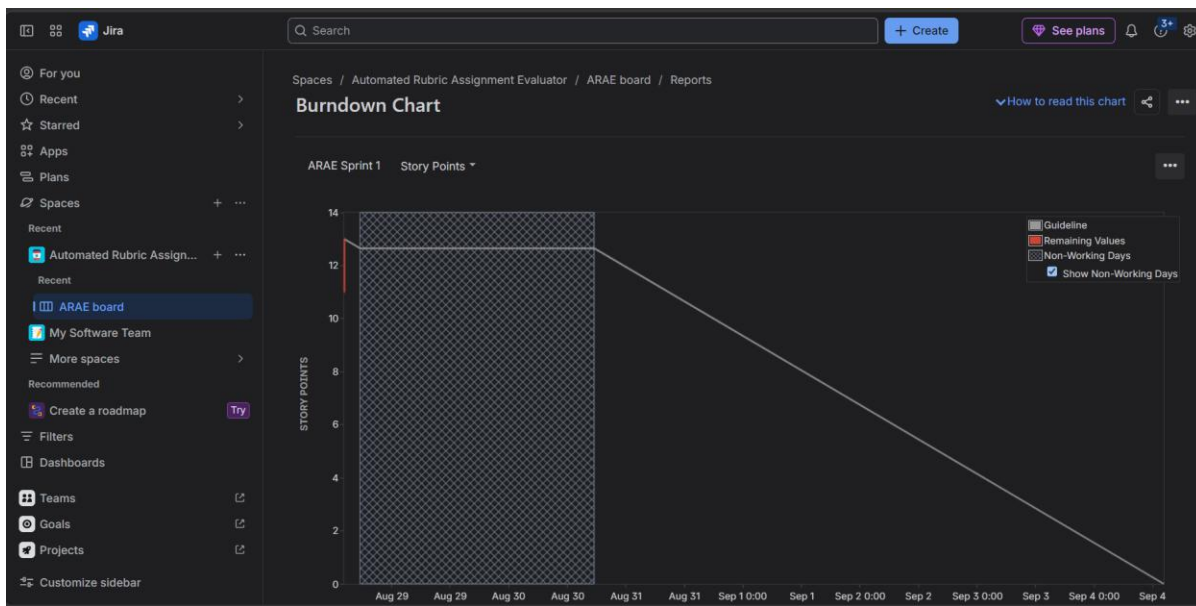
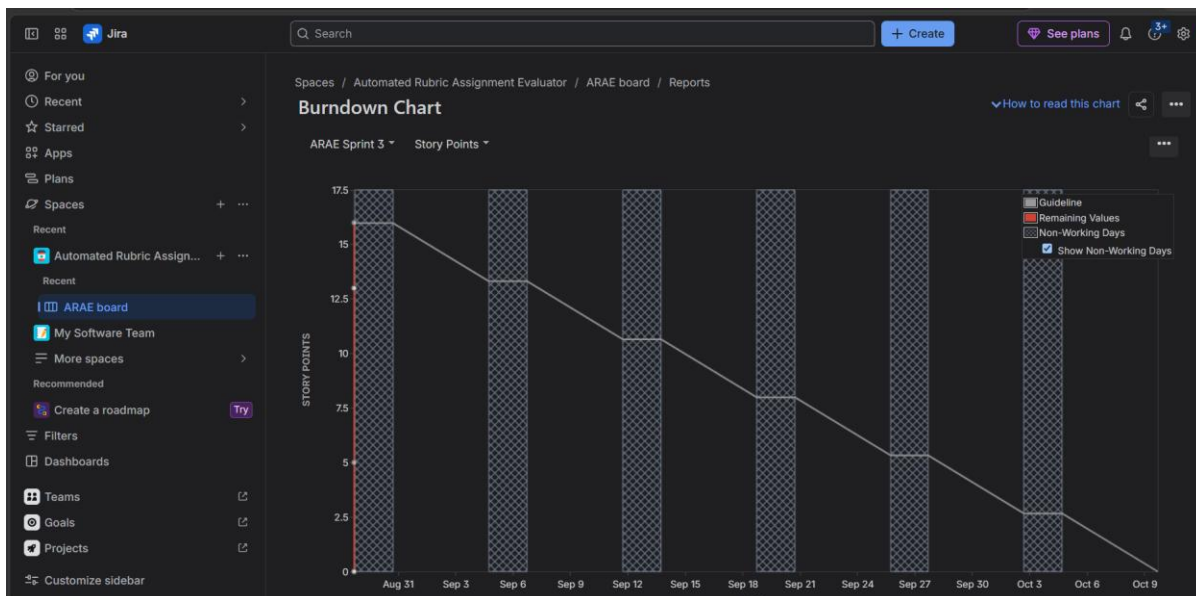


Figure 12: Burndown chart – ARAE Sprint 1 (story points remaining vs. guideline)



7. Reflection Questions

1. Did your estimations reflect the actual effort?

Because this was a simulated sprint, the Story Points were used as relative estimates rather than measured elapsed time. Smaller stories such as Generate Submission ID were assigned fewer points, while more complex stories such as Execute Unit Tests and Evaluate Project Using Rubric received higher points. This gave a reasonable relative estimate of complexity, effort, and uncertainty.

2. Was your backlog well-prioritized?

The backlog was reasonably well-prioritized. Core features such as project submission, validation, syntax checking, unit testing, and rubric evaluation were given higher priority because they support the main functionality of the system. Reporting and peer-review features were scheduled after the core evaluation workflow.

3. How did your simulated sprint align with your plan?

The simulated sprint workflow followed the plan: selected User Stories were moved through the To Do → In Progress → Done stages. The work was organized across the planned sprint iterations, allowing progress to be demonstrated visually on the Jira board.

4. What insights did the burndown chart give about your team's capacity?

The burndown chart showed how the remaining Story Points changed during the sprint compared with the ideal guideline. It provided a visual indication of whether work was being completed steadily and helped assess whether the planned workload was realistic for the sprint.